

Penghui Du

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Research Interests

My current work is on precision functional mapping. I am interested in how individual-specific functional network organization can deepen our understanding of human cognition, neurological / psychiatric disorders, and neuromodulation. I am also interested in integration of multimodal neuroimaging data (such as fMRI, PET, and EEG).

Education

- 2024/08 - 2027/02 **École Polytechnique Fédérale de Lausanne (EPFL)**, *Master in Neuro-X*, GPA: 5.40 / 6.
- 2020/08 - 2024/07 **Southern University of Science and Technology (SUSTech)**.
◦ Bachelor in Intelligent Medical Engineering, GPA: 3.84 / 4, Rank 2 / 22.
- 2023/02 - 2023/06 **University of Zurich (UZH)**, *Visiting Student, Main Subject in Neuroinformatics*.

Research

Harvard University, *Visiting Graduate Student Under Bertarelli Fellowship*.

- 2026/03 - 2027/02 Supervised by Prof. Randy Buckner. Developed iGENMap, a generative method that learns how individual functional eigenmodes correspond to cortical networks, enabling individual-specific network mapping from substantially less fMRI data than current state-of-the-art approaches require.
Investigating precision brain mapping for personalized transcranial magnetic stimulation in depression by comparing empirical, group-level, and individualized targeting strategies and assessing their relative benefits and practical trade-offs.
Max Planck Institute for Human Cognitive and Brain Sciences, *Summer Intern*.
- 2025/06 - 2025/08 Supervised by Prof. Sofie Valk and Dr. Bin Wan in Cognitive Neurogenetics Lab. Developed a deep learning framework to predict individual brain glucose metabolism from structural and functional MRI features, demonstrating that metabolic organization can be reliably inferred from MRI-based representations and generalizes across subjects and datasets.
Athinoula A. Martinos Center for Biomedical Imaging, *Visiting Undergraduate Student*.
- 2023/07 - 2025/12 Supervised by Prof. Jingyuan Chen. Investigated the cortical organization of resting-state metabolic connectivity derived from functional PET-FDG. Identified a robust superior-inferior gradient primarily driven by low-frequency dynamics and aligned with known functional and anatomical organization.
Southern University of Science and Technology, *Undergraduate Research Assistant*.
- 2023/05 - 2023/08 Supervised by Prof. Quanying Liu. We reviewed how large language models can benefit from cognitive sciences, with the goal of integrating cognitive science priors into tests for artificial general intelligence.
- 2022/09 - 2024/07 Supervised by Prof. Quanying Liu. We explored the effects of distinct drugs on the activity and connectivity of brain networks using zebrafish calcium imaging data.

Publications

- 2026 / 08 **Du, P.**, Coursey, S., Xu, T., Jamadar, S., Nolin, S., Wan, B., Wey, H., Polimeni, J., Price, J., Liu, Q., & Chen, J. Human Cerebral Cortex Organization Characterized by Functional PET-FDG "Metabolic Connectivity". *Eur J Nucl Med Mol Imaging* (2026). <https://doi.org/10.1007/s00259-026-08109-5>
- 2026 / 03 **Du, P.** & Chen, J., Data-Driven Methods in Simultaneous EEG-fPET-fMRI: Opportunities and Challenges, at *2025 59th Asilomar Conference on Signals, Systems, and Computers*. <https://doi.org/10.1109/IEEECONF67917.2025.11443781>
- 2024 / 04 Qu, Y., Wei, C., **Du, P.**, Che, W., Zhang, C., Ouyang, W., Bian, Y., Xu, F., Hu, B., Du, K., Wu, H., Liu, J., & Liu, Q. Integration of cognitive tasks into artificial general intelligence test for large models. *iScience*, 2024, 27(4), 109550. <https://doi.org/10.1016/j.isci.2024.109550>

- 2024 / 01 Qu, Y., **Du, P.**, Wei, C., Che, W., Zhang, C., Ouyang, W., Bian, Y., Xu, F., Hu, B., Du, K., Wu, H., Liu, J., & Liu, Q. Promoting interactions between cognitive science and large language models. *The Innovation*, 2024, 5(2).

Posters and Conference Abstracts

- 2026 / 11 **Du, P.**, Du, J., Billot, A., Sun, W., Yeo, B. T. T., Eldaief, M. C., & Buckner, R. L. iGENMap: Individualized Generative Mapping of Cortical Networks Using Functional Eigenmodes. Late-Breaking Abstract (**accepted**), Poster Presentation at *SfN 2026 Annual Meeting*, Washington, DC.
- 2026 / 06 Wan, B., **Du, P.**, Ferrari, A., Saberi, A., John, A., Liu, X., Hansen, J., Liu, Z., Misic, B., Kaiser, S., Thompson, P., Warrior, V., Bethlehem, R., Bernhardt, B., Valk, S. & Kirschner, M., EigenBrain: a spatial framework for reconstructing individual brain molecular architecture from multimodal MRI, Oral Presentation at *OHBM 2026 Annual Meeting*.
- 2024 / 06 **Du, P.**, Coursey, S., Xu, T., Polimeni, J., Wey, H., Liu, Q., & Chen, J. Human Cerebral Cortex Organization Estimated by Functional PET-FDG "Metabolic Connectivity". Poster Presentation at *OHBM 2024 Annual Meeting*.

Awards

- 2025 / 02 EPFL/HMS **Bertarelli Fellowships Program**
- 2024 / 07 SUSTech - **Distinguished Graduate Award**
- 2022 / 10 SUSTech Department of Biomedical Engineering - **Outstanding Student Scholarships**
- 2022 / 06 China College Student Innovation Training Program – **Undergraduate Research Grant**
- 2022 / 06 2022 Guangdong Biomedical Engineering Design Competition – **First Prize**
- 2022 / 05 13th "Challenge Cup" Guangdong Entrepreneurship Plan Competition – **First Prize**
- 2019 / 05 The 13th Asia and Pacific Informatics Olympiad (APIO) – **Bronze Medal**
- 2018 / 11 National Olympiad in Informatics in Henan Province (NOIP) – **First Prize**

Summer Schools

- 2023 / 10 **Laminar fMRI Course**, at *Martinos Center for Biomedical Imaging*.
- 2022 / 08 **Neuromatch Academy Computational Neuroscience Summer School**.
- 2022 / 07 **CLS-CIBR-IDG Summer School in neuroscience**, with *Merit Student Award*.

Talks

- 2026 / 03 **Human Cerebral Cortex Organization Characterized by Functional PET-FDG "Metabolic Connectivity"**, at *Molecular Connectivity Working Group (MCOS) Talk Series*.
- 2024 / 02 **The Organization of Human Cerebral Cortex Estimated by Functional PET-FDG: The Promise and Controversy of Metabolic Connectivity**, at *Science on Tap*.

Teaching experience

- 2022 Spring **JAVA Programming A**, *Teaching Assistant*, SUSTech.
- 2022 Spring **Workshop on Basic Programming**, *Organizer*, SUSTech.

Languages

Chinese: Native speaker.

English: TOEFL score 111 (2024 March).

Programming and Software Skills

Proficient: JAVA, Python, MATLAB, SPM, FSL, LaTeX, Freesurfer, Connectome Workbench.

Working Knowledge: C/C++, AFNI, R, HTML, CSS, Git, Linux, Neurodesk, SimNIBS.